TUTORIAL 3

Fields and Finite Fields

MOTIVATION AND GUIDING IDEA

A proof must work within the given setting: equality of sets is proved by two inclusions, and a field identity is proved from the field axioms. In a finite field, addition, subtraction, multiplication, and division by nonzero elements are permitted; numerical notation does not imply an ordering like that of $\mathbb{R}$.

NOTATION AND CONVENTIONS

The symbol $x \in A$ expresses membership of an element, whereas $A \subseteq B$ expresses inclusion of sets. The quantifiers $\forall$ and $\exists$ mean ``for every'' and ``there exists''; their order matters. The set $A \setminus B$ consists of elements of A absent from B, and $|A|$ is its cardinality when A is finite. For a field F, write $F^\times = F \setminus \{0\}$. The notation $a \equiv b \pmod{p}$ means $p \mid (a-b)$. For prime p, writing $\mathbb{Z}_p = \{0, \dots, p-1\}$ uses representatives of residue classes. The symbol $\mathbb{F}_q$ denotes a field with q elements; when $q = p^n$ with $n > 1$, it must not be identified with arithmetic modulo q.

DEFINITION 3.1

Equality of sets and cancellation. For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. To prove equality, take an arbitrary element from each side and prove that it belongs to the other side. In a field F, for $a, b, c \in F$ with $c \neq 0$,

$$ac = bc \quad \Longleftrightarrow \quad a = b,$$

because multiplication by c^{-1} is permitted. The condition $c \neq 0$ is essential. The characteristic of a field is the least positive integer p for which $p \cdot 1 = 0$, if such an integer exists; otherwise its characteristic is 0. The characteristic of a finite field is prime.

3.1 Algebraic structure: fields

3.1.1 The field axioms

RECALL AND TOOLS

A field is a set F equipped with two operations, addition and multiplication. For all $a, b, c \in F$, the operations satisfy the following axioms. In particular, $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$ is required:

- **Closure:** if $a, b \in F$, then $a +_{\mathbb{F}} b \in F$ and $a \times_{\mathbb{F}} b \in F$.
- **Commutativity:** $a +_{\mathbb{F}} b = b +_{\mathbb{F}} a$ and $a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a$.

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- **Associativity:** $(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c)$ and $(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c)$.
- **Distributivity:** $a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = a \times_{\mathbb{F}} b +_{\mathbb{F}} a \times_{\mathbb{F}} c$ and $(a +_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} c +_{\mathbb{F}} b \times_{\mathbb{F}} c$.
- **Identity elements:** there is $0_{\mathbb{F}} \in F$ such that $a +_{\mathbb{F}} 0_{\mathbb{F}} = a$ for every $a \in F$, and there is $1_{\mathbb{F}} \in F$ such that $a \times_{\mathbb{F}} 1_{\mathbb{F}} = a$ for every $a \in F$.
- **Additive and multiplicative inverses:** every $a \in F$ has an additive inverse $-a$ with $a +_{\mathbb{F}} (-a) = 0_{\mathbb{F}}$; every nonzero $a \in F$ has a multiplicative inverse a^{-1} with $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXAMPLE 3.1

- **Examples:** $\mathbb{Q}$ (rational numbers), $\mathbb{R}$ (real numbers), and $\mathbb{C}$ (complex numbers) are fields with their usual addition and multiplication.
- **Nonexamples:** $\mathbb{Z}$ and $\mathbb{N}$ are not fields, because most of their elements have no multiplicative inverse; for example, 3 has no inverse in $\mathbb{Z}$.

3.1.2 Notation

A general field is usually denoted by F (sometimes $\mathbb{F}$). When no particular field is intended, we write ``Let F be a field.''
Notation convention: when the field is clear from context, we omit its subscript from the operations and the elements $0, 1$.

EXERCISE 3.1

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that $(a^{-1})^{-1} = a$.

WORKED SOLUTION

Recall that every invertible field element has a unique multiplicative inverse, and in particular $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$. Put $x = (a^{-1})^{-1}$ and calculate:

$$(a^{-1}) \times_{\mathbb{F}} x = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a$$

$$a \times_{\mathbb{F}} (a^{-1}) \times_{\mathbb{F}} x = a \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} x = a \qquad (*)$$

Thus $x = a$, that is, $(a^{-1})^{-1} = a$.
Remark: step $(*)$ uses $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

And we're done. Victory!

EXERCISE 3.2

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that the multiplicative inverse of a is unique.

WORKED SOLUTION

Suppose $b \in F$ satisfies $a \times_{\mathbb{F}} b = 1_{\mathbb{F}}$. Multiply on the left by a^{-1}:

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a^{-1}$$

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EXERCISE 3.2 CONTINUED

$$(a^{-1}) \times_{\mathbb{F}} a \times_{\mathbb{F}} b = a^{-1} \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} b = a^{-1} \qquad (*)$$

Hence $b = a^{-1}$, so the inverse of a is unique. *Remark:* again, step $(*)$ uses $(a^{-1}) \times_{\mathbb{F}} a = 1_{\mathbb{F}}$.

You know what that means? V-I-C-T-O-R-Y!

3.2 Further exercises on fields

EXERCISE 3.3

Let a, b be elements of a field $\mathbb{F}$. Please prove that if $a \times_{\mathbb{F}} b = 0_{\mathbb{F}}$, then $a = 0_{\mathbb{F}}$ or $b = 0_{\mathbb{F}}$.

WORKED SOLUTION

We distinguish two cases according to b. **Case 1:** $b = 0_{\mathbb{F}}$, and the conclusion already holds. **Case 2:** $b \neq 0_{\mathbb{F}}$. Since $\mathbb{F}$ is a field, b has an inverse b^{-1}. Multiplying both sides by b^{-1} gives

$$a \times_{\mathbb{F}} b \times_{\mathbb{F}} b^{-1} = 0_{\mathbb{F}} \times_{\mathbb{F}} b^{-1} \Longrightarrow a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 0_{\mathbb{F}}$$

Another one down. Victory!

PROPOSITION 3.1

A field has no nontrivial zero divisors.

EXERCISE 3.4

Let F be a field. On F^3, define

$$(a, b, c) \oplus (d, e, f) = (a + d, b + e, c + f),$$

$$(a, b, c) \otimes (d, e, f) = (ad, be, cf).$$

Please determine whether this is a field.

WORKED SOLUTION

The zero element is $(0, 0, 0)$, but

$$(1, 0, 0) \otimes (0, 1, 0) = (0, 0, 0).$$

Neither factor is zero, since $1 \neq 0$ in F. A field has no nontrivial zero divisors, so this structure is not a field.

We've got it. Victory!

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3.3 Congruence and Modular Arithmetic

DEFINITION 3.2

Write $\mathbb{Z}$ for the integers. For an integer $m \geq 2$, the notation

$$a \equiv b \pmod{m}$$

means that $a - b$ is divisible by m: there is $k \in \mathbb{Z}$ such that $a - b = km$. The residue class of a is the set

$$\overline{a} = \{a + km : k \in \mathbb{Z}\}.$$

Any two elements in the same class are congruent modulo m. There are exactly m classes, represented by $0, 1, \ldots, m-1$.

EXAMPLE 3.2

Modulo 13, we have $-5 \equiv 8$ and $34 \equiv 8$. For example,

$$(-5) + 34 \equiv 8 + 8 \equiv 3, \qquad (-5) \cdot 34 \equiv 8 \cdot 8 \equiv 12 \pmod{13}.$$

PROPOSITION 3.2

Addition and multiplication of classes can be computed using representatives:

$$\overline{a} + \overline{b} = \overline{a+b}, \qquad \overline{a}\,\overline{b} = \overline{ab}.$$

The results do not depend on the representatives. Indeed, if $a' = a + km$ and $b' = b + \ell m$, then

$$(a' + b') - (a + b) = (k + \ell)m,$$
$$a'b' - ab = m(kb + \ell a + k\ell m).$$

Both differences are divisible by m.

NOTATION AND CONVENTIONS

For a prime p, the classes with these operations form the field $\mathbb{Z}_p$, also denoted $\mathbb{F}_p$. We normally omit the bars and use the representatives $0, 1, \ldots, p-1$. Addition and multiplication are modulo p. Division by $a \neq 0$ means multiplication by its inverse a^{-1}, which satisfies $aa^{-1} = 1$ in the field.

PROPOSITION 3.3

Every nonzero $a \in \mathbb{Z}_p$ has an inverse. A prime dividing a product divides at least one factor. Thus

$$ax = ay \implies a(x - y) = 0 \implies x = y.$$

Multiplication by a therefore sends the $p-1$ nonzero elements to $p-1$ distinct nonzero elements. One of them is 1, so there is x satisfying $ax = 1$. The same cancellation argument gives uniqueness. The other field axioms follow from the laws of integer arithmetic.

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NOTE

For example, $5^{-1} = 8$ in $\mathbb{Z}_{13}$ because $5 \cdot 8 = 40 \equiv 1$. Hence $5x = 7$ gives $x = 8 \cdot 7 = 56 \equiv 4$. Primality matters: modulo 4, we have $2 \cdot 2 = 0$ although neither factor is zero. These operations do not form a field. There are also finite fields whose number of elements is not prime; they are not obtained simply by integer arithmetic modulo that number.

EXERCISE 3.5

Let p be an odd prime. Please prove that every $t \in \mathbb{Z}_p$ can be written as $t = x^2 + y^2$. Please find such an expression for 7 in $\mathbb{Z}_{13}$.

WORKED SOLUTION

Let $Q = \{x^2 : x \in \mathbb{Z}_p\}$ be the set of squares. The notation $|Q|$ is the number of its distinct elements. If $x^2 = y^2$, then

$$(x - y)(x + y) = 0.$$

A field has no zero divisors, so $x = y$ or $x = -y$. For $x \neq 0$, the elements $x, -x$ are distinct: equality would give $2x = 0$, and odd p would imply $x = 0$. Thus the $p - 1$ nonzero elements give exactly $(p - 1)/2$ distinct squares. Including 0 gives

$$|Q| = \frac{p+1}{2}.$$

The set $t - Q = \{t - q : q \in Q\}$ has the same number of elements. If Q and $t - Q$ were disjoint, their union would contain $p + 1$ elements inside a set of size p, a contradiction. Their intersection therefore contains an element: there are x, y such that $x^2 = t - y^2$, giving $t = x^2 + y^2$. In the example,

$$4^2 + 2^2 = 16 + 4 = 20 \equiv 7 \pmod{13}.$$

That does it. Victory!

EXERCISE 3.6

Let p be an odd prime, and define $h(x) = x + x^{-1}$ for $x \in \mathbb{Z}_p \setminus \{0\}$. Please prove that h takes exactly $(p + 1)/2$ distinct values. Please find the set of values when $p = 11$.

WORKED SOLUTION

For nonzero x, y,

$$h(x) - h(y) = x - y + \frac{y - x}{xy} = \frac{(x - y)(xy - 1)}{xy}.$$

Division by xy is allowed because both factors are nonzero. Consequently,

$$h(x) = h(y) \quad \Longleftrightarrow \quad y = x \text{ or } y = x^{-1}.$$

The elements equal to their own inverse satisfy $x^2 = 1$, so they are exactly $1, -1$. They are distinct because p is odd. Their values are also distinct: $h(1) = 2$ and $h(-1) = -2$. The remaining $p - 3$ elements split into pairs $\{x, x^{-1}\}$, each giving a value not obtained from

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EXERCISE 3.6 CONTINUED

another pair. The number of values is therefore

$$2 + \frac{p-3}{2} = \frac{p+1}{2}.$$

In $\mathbb{Z}_{11}$, the pairs and singletons are

$$\{1\},\ \{10\},\ \{2,6\},\ \{3,4\},\ \{5,9\},\ \{7,8\}.$$

Their respective values of $x + x^{-1}$ are $2, 9, 8, 7, 3, 4$, so the set of values is

$$\{2,3,4,7,8,9\}.$$

All wrapped up. Victory!

3.4 Finite fields

NOTE

For a prime p, we write $\mathbb{Z}_p = \{0, \ldots, p-1\}$ and perform addition and multiplication modulo p. These integers represent residue classes. The same field is also denoted $\mathbb{F}_p$. Most finite-field examples here use these residue fields.

Definition and notation

DEFINITION 3.3

- A **finite field** is a field with finitely many elements. Write $|\mathbb{F}| = q$.
- Every finite field has $q = p^n$ elements, where p is prime and $n \geq 1$ is an integer. This is a structural theorem about finite fields.
- For a prime p, the set $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, with addition and multiplication modulo p, is a field. It is called the **prime field of characteristic** p.
- Addition and multiplication in $\mathbb{Z}_p$ are defined by

$$a +_{\mathbb{Z}_p} b := (a+b) \bmod p, \quad a \times_{\mathbb{Z}_p} b := (a \cdot b) \bmod p.$$

Examples

EXAMPLE 3.3

- $\mathbb{Z}_2 = \{0, 1\}$, with $1 + 1 = 0$ and $1 \cdot 1 = 1$.
- $\mathbb{Z}_7 = \{0, 1, \ldots, 6\}$, with arithmetic modulo 7; for example, $5 + 4 = 2$ and $3 \cdot 5 = 1$ in the field.
- $\mathbb{Z}_{11} = \{0, 1, \ldots, 10\}$, with arithmetic modulo 11; for example, $8 + 7 = 4$ and $4 \cdot 3 = 1$ in the field.

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Basic properties

PROPOSITION 3.4

In every finite field $\mathbb{F}_q$, the set $\mathbb{F}_q^\times$ of nonzero elements is closed under multiplication, and each of its elements has a unique inverse.

PROPOSITION 3.5

In every finite field $\mathbb{F}_q$, each $a \in \mathbb{F}_q$ satisfies

$$a^q = a.$$

PROPOSITION 3.6

Every nonzero a in a finite field $\mathbb{F}_q$ satisfies $a^{q-1} = 1$.

Explanation: Closure follows because a field has no zero divisors. The existence and uniqueness of inverses follow from the field axioms and the proof above. To prove both power identities, let $a \neq 0$ and put

$$P = \prod_{x \in \mathbb{F}_q^\times} x \neq 0.$$

Multiplication by a permutes the nonzero elements: the map $x \mapsto ax$ is invertible, with inverse $x \mapsto a^{-1}x$. Thus

$$P = \prod_{x \in \mathbb{F}_q^\times} (ax) = a^{q-1}P.$$

Since $P \neq 0$, cancellation is permitted and gives $a^{q-1} = 1$, hence $a^q = a$. For $a = 0$, the identity $0^q = 0$ is immediate.

EXAMPLE 3.4

The nonzero inverses in $\mathbb{Z}_{11}$ are

a	1	2	3	4	5	6	7	8	9	10
a^{-1}	1	6	4	3	9	2	8	7	5	10

For example, $2 \cdot 6 = 12 \equiv 1$, $5 \cdot 9 = 45 \equiv 1$ and $7 \cdot 8 = 56 \equiv 1 \pmod{11}$.

EXERCISE 3.7

Please show that, in every finite field $\mathbb{F}_q$,

$$(a - b)^q = a^q - b^q.$$

WORKED SOLUTION

Use the identity $x^q = x$, proved for every $x \in \mathbb{F}_q$. In particular, it applies to $a - b$, so

$$(a - b)^q = a - b = a^q - b^q.$$

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EXERCISE 3.7 CONTINUED

This proves the claim directly, without calculating binomial coefficients.

We've reached the finish. Victory!

EXERCISE 3.8

Please work in $\mathbb{Z}_{17}$.

1. Please find the inverses of $2, 3, 5, 7, 8$.
2. Please calculate $E = (7^{-1} + 3)(5 - 2^{-1})^{-1} - 2$.

WORKED SOLUTION

All arithmetic is modulo 17. We have

$$2 \cdot 9 = 18 \equiv 1, \quad 3 \cdot 6 = 18 \equiv 1, \quad 5 \cdot 7 = 35 \equiv 1,$$

$$7 \cdot 5 = 35 \equiv 1, \qquad 8 \cdot 15 = 120 \equiv 1.$$

Thus the inverses, in order, are $9, 6, 7, 5, 15$. Now $7^{-1} + 3 = 5 + 3 = 8$, while

$$5 - 2^{-1} = 5 - 9 = -4 \equiv 13 \neq 0.$$

The inverse of 13 is 4, because $13 \cdot 4 = 52 \equiv 1$. Therefore

$$E = 8 \cdot 4 - 2 = 30 \equiv 13 \pmod{17}.$$

Yes! Victory!

EXERCISE 3.9

Please show that for every nonzero a in a finite field $\mathbb{F}_q$, there is $n \in \mathbb{N}$ with $a^n = 1$.

WORKED SOLUTION

The set $\mathbb{F}_q^\times$ contains exactly $q - 1$ elements and is closed under multiplication. Put $m = q - 1$ and consider the list

$$a, a^2, a^3, \ldots, a^{m+1}.$$

It has $m + 1$ entries but only m possible distinct values, so two entries are equal, say $a^k = a^l$ with $k > l$. Multiplying by a^{-l} gives $a^{k-l} = 1$. Thus every nonzero element has a positive integer power equal to the multiplicative identity.

V-I-C-T-O-R-Y!

3.5 Calculations and applications in finite fields

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EXERCISE 3.10

Let $a, b, c, d \in \mathbb{Z}_2$ be four bits. Add a bit e such that $a + b + c + d + e = 0$. Please find e and please calculate it for $(a, b, c, d) = (1, 0, 1, 1)$. Please determine whether the check detects one flipped bit and what happens with two flipped bits.

WORKED SOLUTION

In $\mathbb{Z}_2$, $-x = x$, so $e = a + b + c + d$. For the given example, $e = 1 + 0 + 1 + 1 = 1$. Write the five error bits as $\varepsilon_1, \ldots, \varepsilon_5$, with 1 indicating a flip and 0 no flip. The received parity is

$$(a + b + c + d + e) + \sum_{j=1}^{5} \varepsilon_j = \sum_{j=1}^{5} \varepsilon_j.$$

A single flip gives parity 1 and is detected. Two flips give $1 + 1 = 0$ and are not detected. The check does not identify which bit flipped.

Victory!

Additional practice

EXERCISE 3.11 FOUNDATION

In $\mathbb{Z}_{11}$, define

$$A = \{x \in \mathbb{Z}_{11} : 3x + 5 = 2\}, \qquad B = \{x \in \mathbb{Z}_{11} : x^2 = 1\}.$$

Please find $A, B, A \cup B, A \cap B$ and please determine whether $A \subseteq B$.

WORKED SOLUTION

All arithmetic is modulo 11. The first equation gives $3x = 8$. Since $3^{-1} = 4$, we get $x = 4 \cdot 8 = 32 = 10$, so $A = \{10\}$. The second equation is equivalent to $(x - 1)(x + 1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1 = 10$. Hence

$$B = \{1, 10\}, \quad A \subseteq B, \quad A \cup B = \{1, 10\}, \quad A \cap B = \{10\}.$$

And we're done. Victory!

EXERCISE 3.12 CHALLENGE

Let F be a finite field with $q > 2$ elements. Please prove that

$$\sum_{x \in F} x = 0, \qquad \sum_{x \in F^\times} x^{-1} = 0.$$

Please check both statements in the field with two elements. Do not assume an ordering on F.

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EXERCISE 3.12 CHALLENGE CONTINUED

WORKED SOLUTION

Put $S = \sum_{x \in F} x$. Since $q > 2$, choose $c \in F \setminus \{0, 1\}$. The map $x \mapsto cx$ is bijective, with inverse $x \mapsto c^{-1}x$. It therefore only rearranges the summands, and

$$cS = \sum_{x \in F} cx = \sum_{x \in F} x = S.$$

Thus $(c-1)S = 0$. Since $c - 1 \neq 0$, field cancellation gives $S = 0$. Removing the summand 0 does not change the sum, so $\sum_{x \in F^\times} x = 0$. The map $x \mapsto x^{-1}$ also permutes $F^\times$, because applying it twice returns x. Therefore

$$\sum_{x \in F^\times} x^{-1} = \sum_{x \in F^\times} x = 0.$$

In $\mathbb{Z}_2$, we have $0 + 1 = 1 \neq 0$, and the sum of the nonzero inverses is $1^{-1} = 1 \neq 0$. Indeed, no element c distinct from $0, 1$ exists there, so the choice used in the proof is unavailable.

You know what that means? V-I-C-T-O-R-Y!

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EXERCISE 3.13 CHALLENGE

Let F be a field satisfying $x^3 = x$ for every $x \in F$. Please prove that F has either two or three elements. Please write its operations in each case and please identify them with those of $\mathbb{Z}_2$ or $\mathbb{Z}_3$.

WORKED SOLUTION

For $x \neq 0$, cancellation gives $x^2 = 1$, hence $(x-1)(x+1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1$. Thus $F \subseteq \{0, 1, -1\}$. If $1 = -1$, then $F = \{0, 1\}$ and $1 + 1 = 0$. Together with multiplication by zero and $1 \cdot 1 = 1$, these are the operations of $\mathbb{Z}_2$. If $1 \neq -1$, all three elements are distinct and present. The sum $1 + 1$ cannot be 0, which would give $1 = -1$, or 1, which would give $1 = 0$ by cancellation. Therefore $1 + 1 = -1$. Writing $2 = -1$, we obtain

$$1 + 2 = 0, \qquad 2 + 2 = 1, \qquad 2 \cdot 2 = 1.$$

These, together with the zero and identity rules, give exactly $\mathbb{Z}_3$. Conversely, direct substitution of $0, 1$ in $\mathbb{Z}_2$, and $0, 1, -1$ in $\mathbb{Z}_3$, verifies $x^3 = x$.

Another one down. Victory!

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EXERCISE 3.14 CHALLENGE

Let F be a finite field. Please prove

$$\prod_{x \in F \setminus \{0\}} x = -1.$$

Here $\prod$ multiplies all nonzero elements. Please also prove that, for every $b \in F$,

$$\prod_{a \in F \setminus \{b\}} (b - a) = -1.$$

WORKED SOLUTION

Pair each nonzero x with x^{-1}, except when the two are equal. Each pair contributes 1 to the product. The unpaired elements satisfy $x = x^{-1}$, equivalently

$$x^2 = 1, \qquad (x-1)(x+1) = 0.$$

Since a field has no zero divisors, these are exactly 1 and -1. If they are distinct, their product is -1. If $1 = -1$, only the element 1 remains unpaired, and its value is again -1. This proves the first identity in every characteristic. For fixed b, the values $b - a$ as a ranges over $F \setminus \{b\}$ run through every nonzero field element exactly once. The second product is therefore the same product in a different order.

We've got it. Victory!

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